



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Dr Sukanya Ledalla
Student Name	Billakanti Siri
Roll Number	2520030601
Branch	CSE
Course Outcome	CO2 – B+ Tree with Composite Index

Case Study Topic: MongoDB Compound-Index B+ Tree on (user_id, timestamp)

Work Done Summary:

In this case study, we implemented and analyzed a composite-key B+ Tree index used in MongoDB on (user_id, timestamp). The implementation demonstrates how composite indexes support efficient searching using prefix keys and range queries. The system stores records sorted first by user_id and then by timestamp to improve query performance for large-scale databases.

Implementation Details:

1. Created a composite index structure using Java TreeMap and TreeSet.
2. Implemented insertion of records containing user_id and timestamp.
3. Demonstrated prefix-based indexing behavior of composite B+ trees..
4. Compared efficient prefix search with inefficient suffix-only search.
5. Analyzed the standard B+ tree cost model:

Result: Screenshots/output

```
siribillakanti@siris-MacBook-Air VSC DSA II % javac assignment_CO2.java
java assignment_CO2

===== QUERY 1 : user_id + timestamp range =====
User ID : 42 | Timestamp : 2024-01-01
User ID : 42 | Timestamp : 2024-01-10
User ID : 42 | Timestamp : 2024-01-25

===== QUERY 2 : user_id only =====
User ID : 42 | Timestamp : 2024-01-01
User ID : 42 | Timestamp : 2024-01-10
User ID : 42 | Timestamp : 2024-01-25

===== QUERY 3 : timestamp only =====
User ID : 55 | Timestamp : 2024-01-20
```

GitHub Link:

https://github.com/siribillakanti/DSA2_ASSIGNMENTS/blob/main/assignment_CO2.java